ET AI HACKATHON 2.0

FRAUD NETWORK
GRAPH INTELLIGENCE

An Unsupervised Graph-Based System for Detecting Coordinated Fraud Networks and Money-Mule Activity

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Project Area: AI for Digital Public Safety — Defeating Counterfeiting, Fraud & Digital Arrest Scams

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Live MVP: kan9667-fraud-network-graph-intelligence-app-59zwzv.streamlit.app

Source: github.com/kan9667/fraud-network-graph-intelligence

1. Executive Summary

Financial fraud is increasingly coordinated. Rather than a single suspicious account or an isolated transaction, organized fraud networks typically involve multiple mule accounts, shared devices, reused identities, coordinated money movement, and rapid transfers across accounts.

Traditional fraud detection systems often focus on individual transactions or rely heavily on predefined rules and labeled historical fraud data, which makes it difficult to identify new or evolving fraud patterns.

Fraud Network Graph Intelligence is an unsupervised fraud detection and investigation system designed to identify coordinated fraud networks from account, identity, device, and transaction data.

The system models financial activity as a graph in which accounts are nodes and relationships such as transactions, shared devices, shared phone numbers, and shared IP addresses are weighted edges. Community detection is then used to automatically discover tightly connected groups without requiring fraud labels during detection.

Discovered communities are evaluated with a multi-factor risk scoring model spanning network structure, identity reuse, behavioral patterns, money concentration, and temporal signals. Suspicious networks are enriched with explainable evidence, inferred behavioral roles, money-flow paths, and timeline analysis.

The system is delivered through an investigator-oriented Streamlit Command Center that guides users through:

Network Discovery → Risk Assessment → Network Investigation → Role Inference → Money-Flow Analysis → Timeline Analysis → Evidence Report

The goal is not simply to produce a fraud score — it is to help investigators understand how a suspicious network operates and why it was flagged.

2. Problem Statement

Fraud networks are difficult to detect because their activity is distributed across multiple accounts and transactions. A coordinated fraud operation may involve:

- Multiple accounts controlled by the same individual or group
- Reuse of phone numbers or devices
- Partial identity or infrastructure overlap
- Money-mule accounts used to move funds
- Coordinators connecting different parts of the network
- Consolidators collecting funds from multiple accounts
- Cash-out accounts transferring funds toward exit destinations
- Rapid forwarding of received money
- Legitimate-looking transactions mixed with suspicious activity

A conventional account-level approach investigates each account separately and misses the relationships connecting them. The key challenge is therefore:

> ***How can we automatically discover coordinated fraud networks from transaction and account data — without relying on predefined fraud labels or manually written rules — and provide investigators with explainable evidence for further investigation?***

3. Proposed Solution

Our solution uses graph intelligence and unsupervised community detection to identify suspicious groups of accounts. The system consists of the following stages.

Step 1 — Data Generation and Ingestion

The MVP uses a deterministic synthetic dataset containing 247 accounts, 937 transactions, 4 synthetic fraud networks, and 200 normal accounts. The dataset intentionally spans different fraud scenarios to test the robustness of the detection approach:

- **Classic money-mule network —** shared phone and device signals; coordinated flow from victims through mules toward cash-out destinations.
- **Device reuse ring —** shared device signals with unique phone numbers, testing detection without phone reuse.
- **Low-identity-signal network —** unique phones and devices, relying primarily on behavioral and transaction patterns.
- **High-noise fraud network —** partial device/IP reuse with heavy noise and interaction with normal accounts, simulating a harder real-world scenario.

Ground-truth labels are maintained separately and used only for offline evaluation — the detection pipeline never accesses them.

4. Graph-Based Detection Architecture

The core of the system is a financial relationship graph.

Nodes

Each node represents an account.

Edges

Edges represent relationships between accounts, including financial transactions, shared phone numbers, shared device IDs, and shared IP signals where available. Identity relationships receive higher graph weight than ordinary transaction relationships, since shared infrastructure can indicate coordinated account control.

The graph is serialized and reused across the pipeline to ensure graph construction has a single source of truth.

5. Unsupervised Community Detection

Once the graph is constructed, the system applies Louvain community detection to identify tightly connected groups of accounts. The system is never told which accounts are fraudulent — instead, the algorithm discovers communities from the structure of relationships alone.

Each detected community is analyzed independently, which allows the system to identify potential fraud networks even when individual accounts do not appear suspicious in isolation. The resulting clusters are ranked using a multi-factor risk scoring system, described next.

6. Multi-Factor Risk Scoring

A major improvement over a simple binary fraud rule is the use of a multi-factor risk model. The risk score considers several independent categories of evidence:

Network Structure

- Community size
- Network density
- Internal connectivity
- Average degree
- Relationship concentration

Identity and Infrastructure

- Shared phone numbers
- Shared devices
- Shared IP signals
- Identity reuse patterns

Behavioral Signals

- Internal transaction concentration
- Money-flow concentration
- External counterparties
- Account interaction patterns

Temporal Signals

- Rapid forwarding
- Transaction timing
- Inbound-to-outbound delays
- Burst-like activity

Money Flow

- Internal transaction volume
- External inbound volume
- External outbound volume
- Exit volume
- Potential forwarding paths

The final risk score is normalized to a 0–100 scale and categorized into LOW, MEDIUM, HIGH, and CRITICAL risk levels — allowing the system to distinguish a normal connected community from one exhibiting multiple coordinated fraud indicators.

7. Behavioral Role Inference

The system goes beyond identifying suspicious clusters. Within flagged networks, it infers potential behavioral roles from transaction patterns:

- **Suspected Victim —** accounts sending or receiving funds in patterns consistent with being targeted by the network.
- **Probable Mule —** accounts that receive and forward funds through the suspicious network.
- **Probable Coordinator —** accounts that connect multiple parts of the network with central behavioral patterns.
- **Probable Consolidator —** accounts that receive funds from multiple network participants and concentrate money flows.
- **Probable Cash-Out Account —** accounts that receive funds within the network and forward significant amounts toward exit destinations.

These are algorithmic behavioral inferences, not confirmed identities. The system explicitly communicates this distinction throughout the interface to avoid treating automated predictions as factual attribution.

8. Money-Flow Intelligence

The Money Flow module focuses on the movement of funds rather than network structure alone. It identifies internal money movement, external inflows and outflows, potential exit destinations, rapid forwarding events, and multi-hop money-flow paths — for example:

Victim → Mule → Coordinator → Consolidator → Cash-Out → External Destination

This lets an investigator understand the operational structure of a suspicious network. Instead of simply reporting "Account X is suspicious," the system provides context such as: "This account received funds from within the suspicious network and subsequently transferred a significant portion toward external exit destinations within a short time period." This makes the detection output more actionable and explainable.

9. Investigator Command Center

The Streamlit MVP provides a complete, investigator-oriented workflow across seven linked views.

Command Center

A high-level overview of detected cases, risk levels, total accounts and transactions, rapid-forwarding activity, and suspicious transaction volume — with a one-click recommended case to start an investigation.

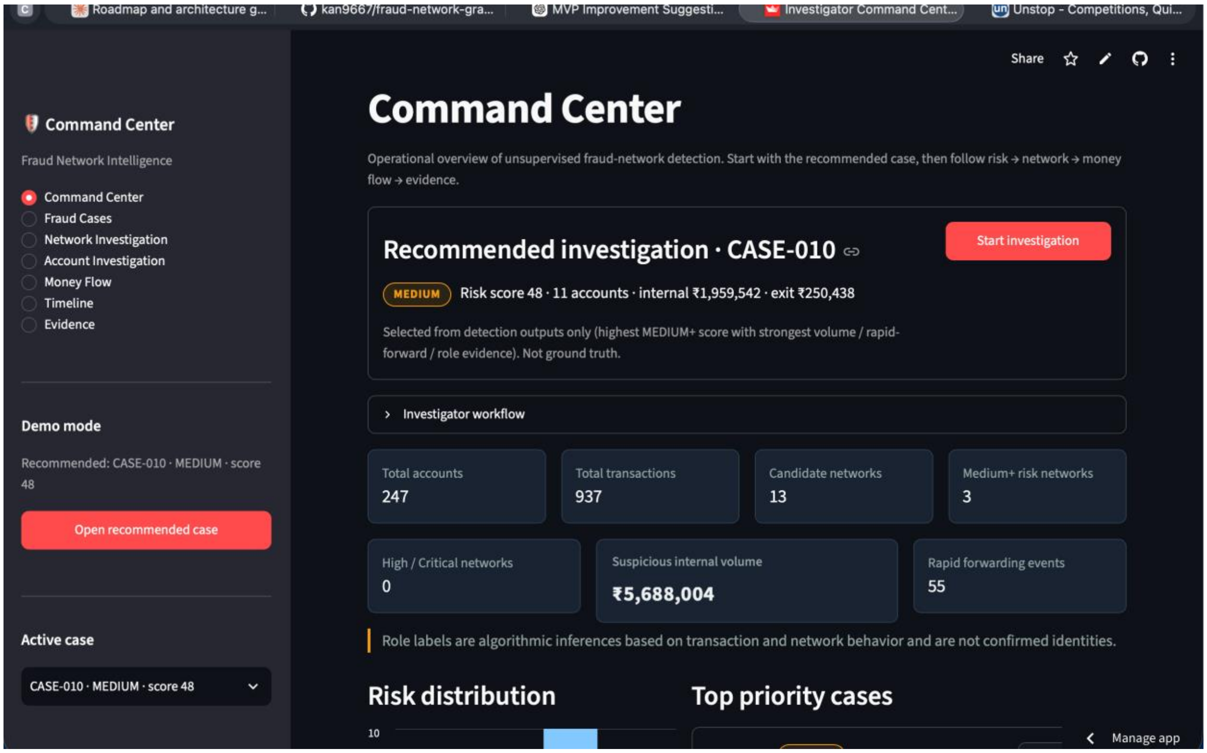

Figure 1 — Command Center: system-wide overview and recommended case

Fraud Cases

A filterable case queue that lets investigators sort and prioritize detected networks by risk level, minimum risk score, and inferred roles present.

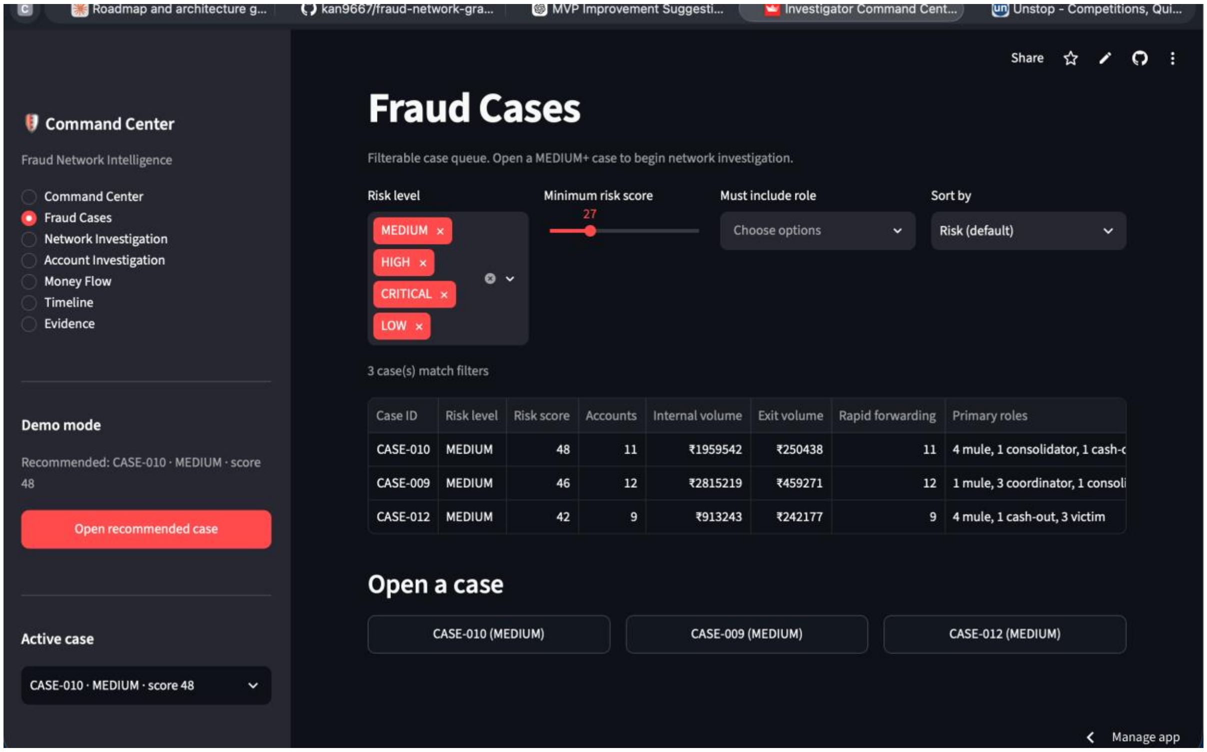

Figure 2 — Fraud Cases: filterable, risk-ranked case queue

Network Investigation

An interactive network graph with account relationships, algorithmically inferred roles, risk-factor explanations, and network structure statistics (density, rapid-forwarding events, internal/exit volume).

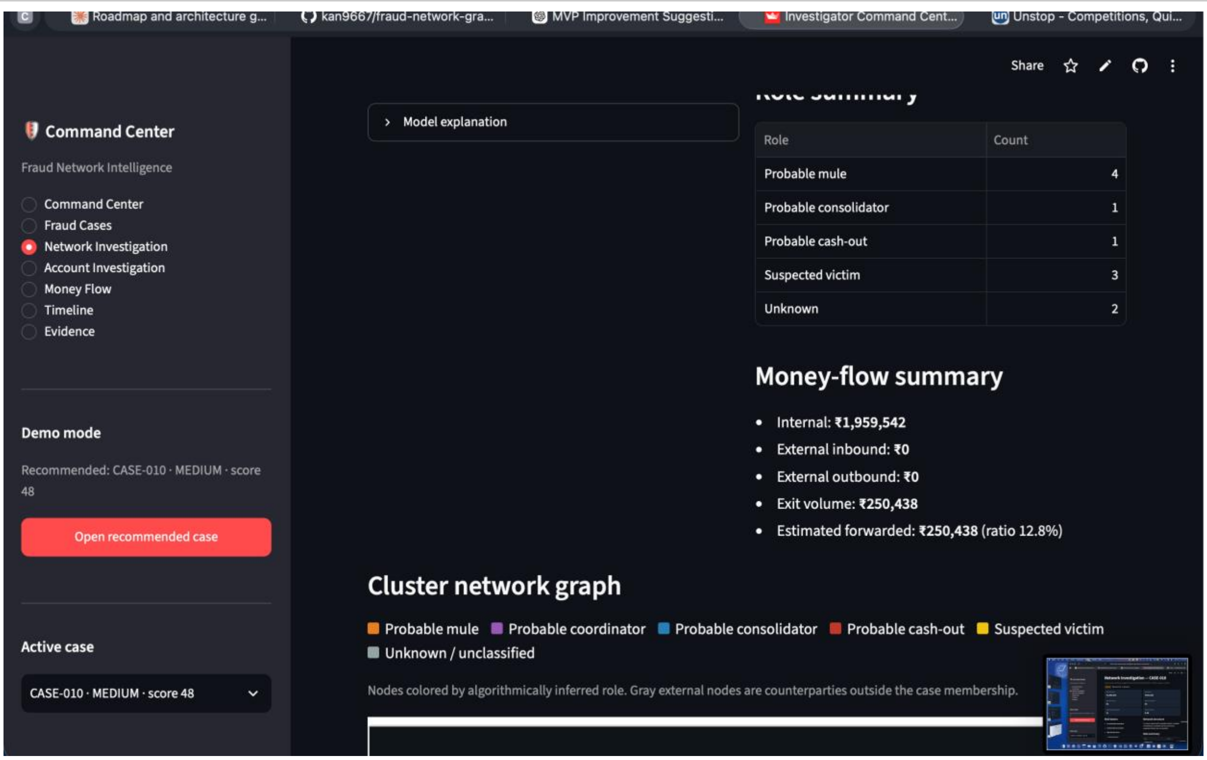

Figure 3 — Network Investigation: risk factors and inferred role summary

Account Investigation

Single-account drill-down showing role evidence, transaction and network statistics, temporal statistics, and the account's immediate network neighborhood.

Money Flow

Internal vs. external volume, exit volume, forwarding ratio, and ranked money-flow paths derived directly from transactions.

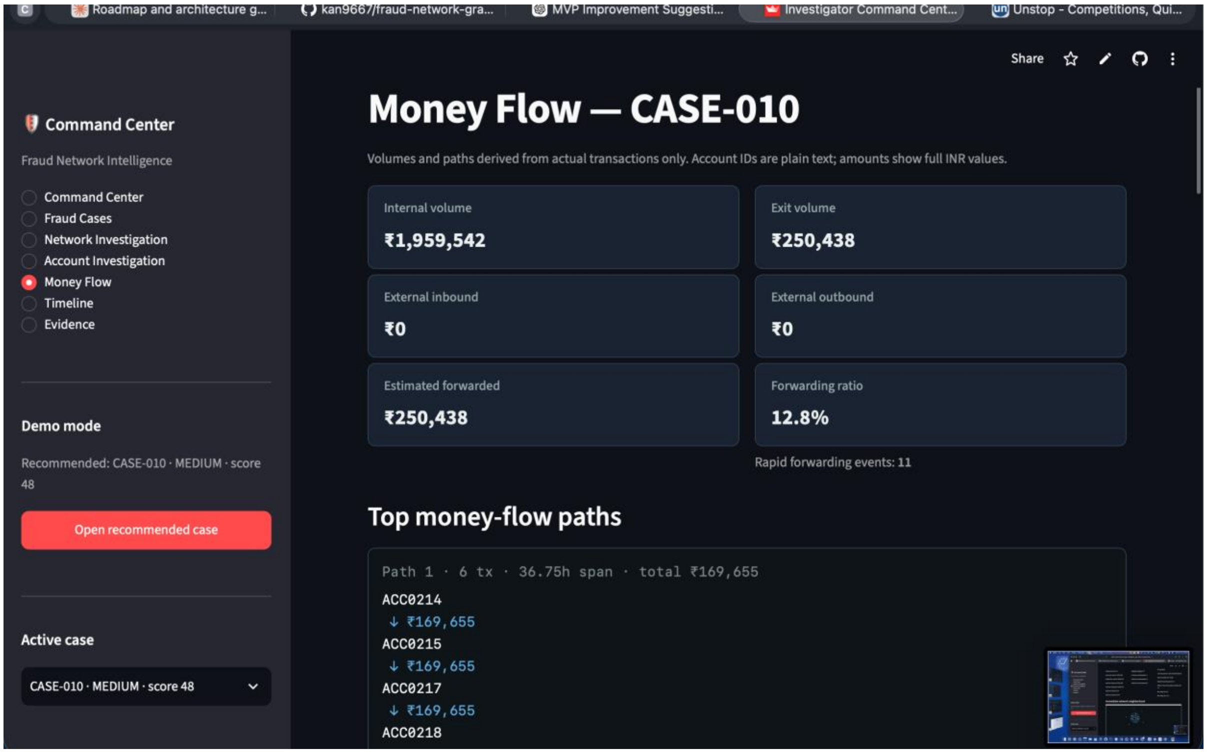

Figure 4 — Money Flow: exit volume, forwarding ratio, and top money-flow paths

Timeline

Chronological transaction view with filters for high-value transfers, rapid forwarding, and exit activity, plus a daily transaction-volume chart.

Evidence

A structured, investigator-ready case package — case summary, risk indicators, per-account role evidence with confidence scores, and one-click export as JSON or Markdown.

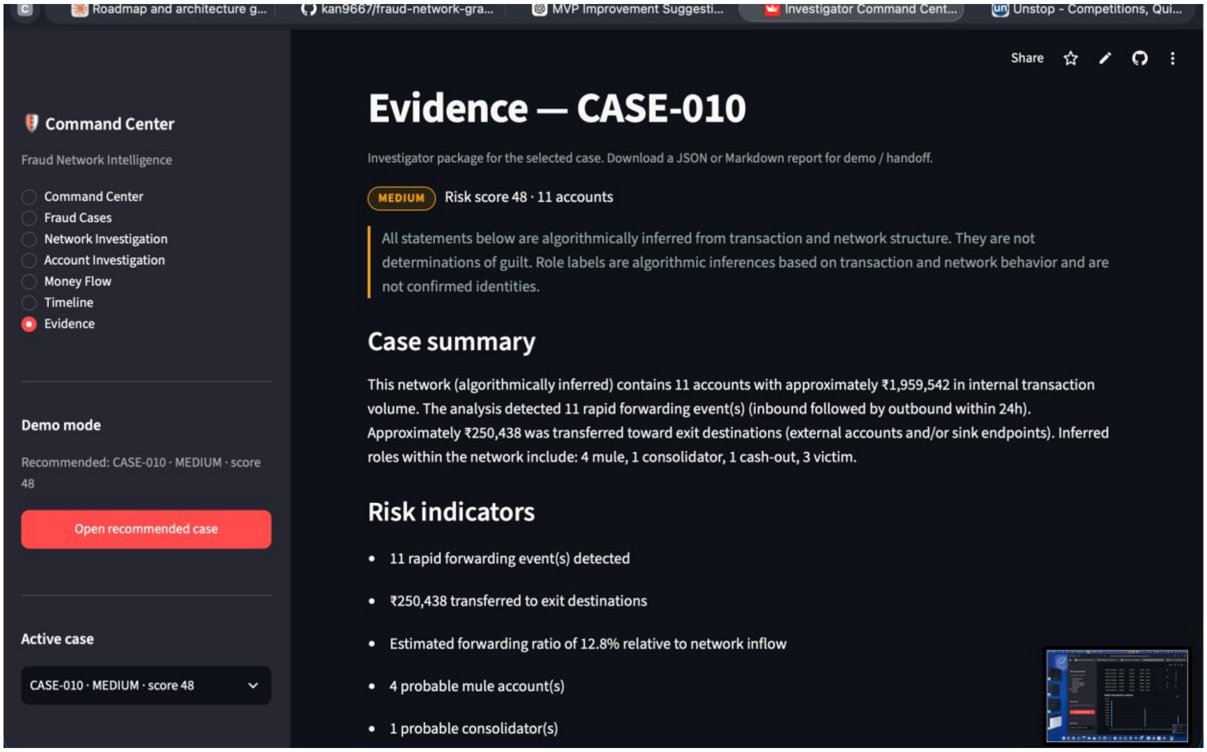

Figure 5 — Evidence: case summary, risk indicators, and exportable report

10. Example Investigation — CASE-010

One representative case generated by the system:

Metric	Result
Risk Level	**MEDIUM**
Risk Score	**48 / 100**
Accounts	**11**
Internal Transaction Volume	**≈ ₹19,59,542**
Potential Exit Volume	**≈ ₹2,50,438**
Rapid Forwarding Events	**11**

The network contains inferred behavioral roles including probable mules, a probable consolidator, a probable cash-out account, suspected victims, and unclassified accounts. The investigation workflow allows an investigator to move from the network graph to money-flow paths and then to the timeline of individual transactions.

An example cash-out pattern identified by the system showed an account that:

- Received approximately ₹1,69,655 from within the suspicious network
- Transferred approximately ₹2,50,438 toward two exit destinations
- Sent roughly 74% of outbound volume outside the active network core
- Had a median inbound-to-outbound delay of approximately one hour

This demonstrates how the system combines structural, behavioral, and temporal evidence to support a real investigation, not just a numeric alert.

11. Evaluation

The system includes a dedicated evaluation pipeline that is completely separated from the detection pipeline. The ground-truth dataset is used only during offline evaluation. Results for MEDIUM+ account detection:

Metric	Result
Precision	**100%**
Recall	**68.09%**
F1 Score	**81.01%**
Accuracy	**93.93%**
True Positives	**32**
False Positives	**0**

Metric	Result
False Negatives	**15**
True Negatives	**200**

The system recovered 3 out of 4 synthetic fraud networks under the current MEDIUM+ risk threshold. The three successfully recovered networks represented different detection challenges, including shared phone/device signals, device reuse, and partial device/IP reuse.

The fourth network revealed an important failure mode: the system correctly identified the underlying community structure, but its risk score remained below the MEDIUM threshold. This demonstrates that community discovery and risk prioritization are separate challenges that must both be tuned.

12. Key Evaluation Insight

One of the most important findings from the evaluation was the strength of behavior-based signals on their own. The low-identity-signal fraud network — which used unique phone numbers and unique devices — was still successfully clustered structurally.

This shows the system is not fundamentally dependent on identity reuse. In an ablation experiment using behavior-only signals, the system was capable of recovering all four synthetic fraud networks. This supports the central design principle:

> ***Fraud networks can be detected from coordinated behavior and financial relationships, even when shared identity or device signals are unavailable.***

This matters for real-world deployment, where fraudsters may deliberately avoid reusing obvious identifiers.

13. Innovation and Differentiation

- **Network-Level Detection —** analyzes relationships between accounts, not accounts in isolation.
- **Unsupervised Discovery —** the detection pipeline requires no labeled fraud examples.
- **Explainable Risk —** cases carry risk factors and evidence, not just a numerical score.
- **Behavioral Role Inference —** identifies the likely operational role of each account within a network.
- **Money-Flow Intelligence —** traces the potential movement of funds through the network.
- **Temporal Investigation —** incorporates rapid forwarding and transaction timing into the analysis.
- **Investigator-Centric Interface —** designed around the workflow of investigating a case, not just displaying a prediction.
- **Ground-Truth Isolation —** evaluation labels are isolated from detection, avoiding leakage and preserving a genuinely unsupervised pipeline.

14. Public Safety Impact

The system is designed to support financial crime investigation and digital public safety teams. Potential applications include:

- Money-mule network discovery
- Coordinated account investigations
- Fraud-ring prioritization
- Financial crime intelligence
- Digital investigation support
- Network-based suspicious activity analysis

The system helps investigators move from a large volume of disconnected transactions to a smaller number of prioritized networks, giving a structured view of who is connected, how money is moving, which accounts appear to play important roles, why a network is suspicious, and what evidence to examine next.

15. Responsible Use

The system is designed as an investigative decision-support tool, not an automated system for declaring individuals guilty of fraud. The following principles guide its use:

- Risk scores represent prioritization, not proof of criminal activity.
- Behavioral roles are algorithmic inferences, not confirmed identities.
- Suspicious accounts require human investigation and additional evidence.
- The system should not be used as the sole basis for enforcement action.
- False positives and false negatives remain possible.
- Human investigators should review the evidence before taking action.

The dashboard explicitly communicates that inferred roles are behavioral predictions rather than confirmed identities, throughout every relevant screen.

16. Current Limitations

The current MVP is a proof of concept built on synthetic data. Key limitations include:

- **Synthetic Dataset —** evaluation uses generated data rather than real-world financial transactions.
- **Risk Calibration —** the current run produces MEDIUM cases but no HIGH or CRITICAL cases.
- **Role Classification —** coverage is limited because roles are intentionally assigned only within MEDIUM+ cases.
- **Temporal Modeling —** temporal signals support investigation but need further work to become a stronger independent detection signal.
- **Graph Fragmentation —** community detection can split a single fraud network into multiple communities, reducing account-level recall.
- **Scalability —** the MVP has not yet been benchmarked against very large financial transaction graphs.

17. Future Roadmap

1. Advanced Temporal Graph Analysis

Develop dynamic graph models that analyze how communities evolve over time.

2. Improved Community Recovery

Investigate overlapping community detection and graph clustering methods to reduce fragmentation of large fraud networks.

3. Better Risk Calibration

Improve the scoring model so high-risk networks receive stronger separation from normal communities.

4. Machine Learning Integration

Introduce optional unsupervised anomaly detection techniques to complement graph-based signals.

5. Real-World Data Integration

Evaluate the system using anonymized or appropriately governed real-world transaction datasets.

6. Scalable Graph Infrastructure

Explore graph databases and distributed processing for larger transaction networks.

7. Advanced Investigation Features

- Cross-case network linking
- Historical account behavior
- Network evolution tracking
- Automated case prioritization
- Investigator feedback loops
- More advanced graph analytics

18. Conclusion

Fraud Network Graph Intelligence demonstrates how graph-based intelligence can transform fraud detection from isolated transaction analysis into network-level investigation. The system automatically discovers communities, scores suspicious behavior, infers potential account roles, traces money movement, analyzes temporal activity, and presents results through an investigator-focused dashboard.

The MVP achieved:

- 100% precision
- 68.09% recall
- 81.01% F1 score
- 93.93% accuracy
- 3 of 4 synthetic fraud networks recovered at MEDIUM+ thresholds

- Zero false positives in the evaluated run

Most importantly, the system demonstrates that coordinated fraud can be identified through behavioral and network relationships, even when obvious identity reuse is absent.

The ultimate objective is to give investigators more than an alert — a network, an explanation, a money trail, and an investigative starting point.

19. Project Links

Live MVP: kan9667-fraud-network-graph-intelligence-app-59zwzv.streamlit.app

Source Code: github.com/kan9667/fraud-network-graph-intelligence

Demo Video: Google Drive — Demo Video